

EMMANUEL BOLARINWA

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EDUCATION

CARNEGIE MELLON UNIVERSITY AFRICA

M.S. Engineering Artificial Intelligence

2026 – 2028

Kigali, Rwanda

Focus: speech and language technology for low-resource languages, artificial intelligence for education, energy and climate modeling.

OBAFEMI AWOLOWO UNIVERSITY

B.Sc. Computer Engineering, Second Class Honours (Upper Division)

2017 – 2023

Ile-Ife, Nigeria

Thesis: low-cost electronic nose for gas classification, using support vector machines, gradient boosting and principal component analysis. Published as a peer-reviewed paper.

RESEARCH EXPERIENCE

LYNGUAL LABS

Artificial Intelligence Researcher

Nov 2024 – Present

Lagos, Nigeria

- Co-led the collection and transcription of 100 hours of Yoruba-English code-switched speech across 10+ domains, creating one of the first open corpora for this language pair.
- Fine-tuned and benchmarked multilingual speech recognition models on code-switched speech, beating monolingual baselines and producing three peer-reviewed papers in 2025.

OPEN ENERGY TRANSITION

Energy Systems Modeler

2023 – Present

Bayreuth, Germany (Remote)

- Built concentrated solar power modeling with thermal storage and a Monte Carlo uncertainty module into the open-source PyPSA-Earth framework; also a maintainer and the documentation lead for the project.
- Ran a comparative geographic analysis of ENTSO-E and OpenStreetMap grid data to move the framework onto fully open data.

PERVASIVE AND MOBILE COMPUTING RESEARCH GROUP

Undergraduate Researcher, supervised by Dr. S. Aina

2023 – 2024

Ile-Ife, Nigeria

- Built the full machine learning pipeline for a low-cost electronic nose, from sensor data collection to gas classification.

PUBLICATIONS

2025 *Closing the Gap in Low-Resource ASR: Leveraging Multilingual Models for Code-Switched Yoruba-English Speech.* Bolarinwa, E., et al. Deep Learning Indaba 2025 (top 1% paper award); to appear in PMLR.

2025 *Beyond Monolingual Limits: Fine-Tuning Monolingual ASR for Yoruba-English Code-Switching.* Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., et al. CALCS Workshop at NAACL 2025.

2025 *Challenging Multimodal LLMs with African Standardized Exams: A Document VQA Evaluation.* Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., et al. AfricaNLP Workshop at ACL 2025.

2024 *A Low-Cost Electronic Nose Device.* Aina, S., et al. International Journal of Information Research and Review, 11(5).

SELECTED PROJECTS

AFRILEARNER360 | *Python, large language models, React*

2026

- Co-built a low-resource system that profiles primary school learners by how they engage with content and adapts curriculum material to local culture; led the profiling work and its integration with content delivery.

YORUBA-ENGLISH CODE-SWITCHING SPEECH CORPUS | *Python, audio processing*

2025

- Curated a code-switched speech dataset through collection, segmentation, language-boundary marking and bilingual transcription.

GITHUB ARCHIVE DATA PIPELINE | *Terraform, Prefect, PySpark, dbt, BigQuery*

2023

- Engineered an end-to-end batch pipeline on Google Cloud, from infrastructure provisioning through transformation to Looker dashboards.

TEACHING, AWARDS AND SKILLS

Teaching Machine Learning Tutor, Lyngual Labs Academy (2025 – present); Invited Lecturer, International Centre for Hydropower workshop, Nairobi (2024); Data Analysis Instructor, IJAN Africa (2023 – 2024); Teaching Assistant, Obafemi Awolowo University (2023 – 2024).

Awards Paper Award, Deep Learning Indaba 2025 (top 1% of submissions); youngest invited lecturer in the history of the International Centre for Hydropower (2024).

Skills Python, SQL, PyTorch, TensorFlow, scikit-learn, Keras, MLflow, Docker, Git, PySpark, dbt, Prefect, Terraform, PyPSA, Snakemake, GeoPandas, Google Cloud, Amazon Web Services. Languages: English, Yoruba.